

Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06) held in
Uttarakhand, India

Brain Tumor Radiogenomic Classification using multilevel contrastive information fusion

Asmita^{a*}, Praveen Mittal^a

^aDepartment of Computer Engineering Application, GLA University, Mathura, Uttar Pradesh 281406, India

Abstract

Brain tumors present significant challenges in diagnosis and treatment due to their complex and diverse nature. Radiogenomic analysis, which combines genomic and radiological data, has become important for advancing brain tumor research, with deep learning methods, particularly fusion-based approaches, making notable progress. However, current approaches face challenges in capturing both sequential and spatial features, as well as selecting effective fusion techniques. The research introduces a multi-fusion framework that combines Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and transformers using contrastive learning to enhance feature representation. This approach processes medical images through a neural network alongside a transformer backbone, merging outputs at various stages to improve dark knowledge differentiation between classes. While CNNs extract spatial features, transformers focus on patch-specific information, collectively boosting the model's discriminative power. The method demonstrates a 2-3% performance improvement over state-of-the-art models, with significant advancements in glioblastoma diagnosis and treatment. An ablation study further confirms that each component positively contributes to the classification task.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06)

Keywords: Brain tumor, Radiogenomic Classification, Deep Learning, Image Classification

1. Introduction

Brain tumors represent a highly complex and heterogeneous group of chronic conditions, posing significant health challenges on a global scale. Despite substantial research efforts focusing on their molecular and genetic characteristics, the accurate diagnosis, prognosis, and treatment of brain tumors continue to be formidable tasks. In

recent years, the integration of genomic information with radiological data, collectively termed as radiogenomic data, has emerged as a key tool in advancing brain tumor analysis. The application of deep learning techniques in this domain has driven notable progress, leading to significant advancements in diagnosis and treatment strategies [4,5,7]. Among various approaches, fusion-based techniques have gained traction, demonstrating considerable promise in enhancing the efficacy of these models [3].

Recent advances in medical data classification have seen the development of fusion-based methods, knowledge distillation, self-supervised and supervised contrastive learning, and hybrid learning approaches [4,5,6,7]. These techniques play a critical role in improving classification tasks in the medical field, particularly in radiogenomic studies. One of the most significant biomarkers in glioblastoma research is MGMT promoter methylation, which has been shown to influence the aggressiveness of glioblastoma, the most common and lethal brain cancer in adults. The RSNA-MICCAI Brain Tumor Radiogenomic Classification Challenge [2] aimed to predict MGMT promoter methylation status using a binary classification model based on multiple types of multiparametric MRI (mpMRI) scans.

Radiogenomic-based imaging techniques offer a less invasive way to classify glioblastoma, potentially improving patient survival and treatment outcomes. Knowing the MGMT promoter methylation status is crucial, as it helps guide therapy—methylated tumors often respond better to specific treatments. In a recent challenge, models were trained on various mpMRI brain scans (T1w, T1wCE, T2w, and FLAIR) to predict the likelihood of glioblastoma. The final diagnosis was made by combining the results from these individual models, providing a confidence score for the tumor classification.

The major challenges in existing approaches arise from the inherent limitations of using CNNs or RNNs individually [1]. CNNs are limited in capturing temporal or sequential information effectively, while RNNs may fail to capture critical spatial discriminative features [16,17,18]. Although fusion techniques offer a potential solution, selecting the most appropriate fusion method remains a significant challenge. Furthermore, while transformers are effective at capturing global information, they often struggle to focus on local features. The issue of data scarcity further complicates fusion, as it may lead to the overemphasis of redundant information.

Based on these, the motivation for this work stems from the need to address challenges in efficiently extracting and utilizing dark knowledge for improved model performance in data-scarce scenarios. The key objective is to develop a robust multilevel fusion framework that leverages CNNs and transformers for complementary feature extraction, employs contrastive learning to enhance representation quality, and integrates intermediate and decision-level fusion to refine predictions. This approach aims to mitigate the fading of dark knowledge, effectively capturing both local and global information, and achieving superior performance compared to existing state-of-the-art methods.

To achieve these objectives and address the above-mentioned challenges, the proposed methodology consists of three key phases. In the first phase, a cascade network is trained by utilizing both a convolutional neural network (CNN) and a transformer in parallel, allowing for complementary feature extraction. The second phase involves intermediate-level fusion, where feature maps are extracted using contrastive learning and subsequently fused to improve representation quality. In the final phase, decision-level fusion is performed by aligning the confidence scores of the logits to refine and finalize the model's predictions.

The major contributions in this article are as follows:

- A multilevel fusion approach is proposed, which more effectively captures both local and global information, mitigating the fading of dark knowledge and enhancing its utilization for downstream tasks.
- Contrastive learning is employed to extract more robust dark knowledge, improving the representation quality for downstream tasks.
- The overall method demonstrates a 2-3% performance improvement over the existing state-of-the-art (SOTA) approaches.

The remained article is organized as follows: Section 2 reviews related works. Section 3 presents the proposed methodology. Section 4 outlines the experimental settings. Section 5 discusses the results. Section 6 concludes with key findings and future directions.

2. Related works

Computer-aided diagnosis (CAD) using deep learning has gained significant momentum in recent years, offering innovative solutions for early disease detection, intelligent treatment strategies, and prognosis [4,5,7,8]. Radiogenomic classification, as part of CAD, plays a critical role in this context, with deep learning-based fusion methods emerging as a key component. The current body of literature on CAD covers a wide range of applications, including skin melanoma classification, diabetic retinopathy detection, and brain tumor identification. These studies employ a variety of techniques, such as fusion-based approaches, synergistic neural networks, self-ensembling regularization, 3D imaging methods, structural alignment, and uncertainty-driven contrastive learning, highlighting the growing importance of deep learning in medical diagnosis.

In the area of CADs, Brain tumor detection using MRI scans has emerged as a critical task in recent years [19,20]. Abdusalomov et al. [9] proposed a deep learning approach leveraging a pre-trained object detection model combined with a Bi-directional Feature Pyramid Network for feature extraction and a Channel and Spatial Attention module to enhance attention mechanisms. While their model delivered strong performance, its emphasis on large tumors and reliance on specific MRI scanners and acquisition protocols highlight the need for further exploration. Saeedi et al. [10] employed data augmentation and pre-processing alongside 2D convolutional deep learning techniques for MRI-based brain tumor detection, yet improvements in generalization to small tumors and robustness across different scanners are necessary. Yogananda et al. [11] introduced a 3D-dense-UNet architecture to determine glioma MGMT promoter methylation status, achieving notable results, although some code-related issues remain unresolved. Additionally, S. A. Qureshi [12] proposed a novel radiogenomic classification system with a two-stage prediction process, delivering excellent performance in predicting MGMT promoter methylation status in glioblastoma patients using a multi-omics fused feature space combined with k-NN and SVM classifiers. Kollias et al. [3] presented methods involving data augmentation, 3D analysis using CNN-RNN for global analysis, routing through a mask layer to handle variable input feature lengths, and modality fusion. Mohamed et al. [1] explored the use of advanced deep neural networks for classification. While these methods show promise, addressing redundant features and improving model robustness remain key challenges for future advancements.

Apart from these, the extraction and application of hybrid learning methods and dark knowledge are pivotal in enhancing the training and regularization of deep neural networks, especially when addressing challenges posed by data scarcity [6,13,14,15]. Advanced techniques such as fusion for improved learning, intermediate knowledge distillation, contrastive learning, and self-supervised dark knowledge extraction significantly optimize the learning capabilities and robustness of neural networks, leading to better performance. These methods play a crucial role in leveraging limited data by maximizing learning outcomes and enhancing model reliability. For similar classification tasks, Qu and Xiao [16] propose an attentive multimodal CNN framework that integrates diverse data types effectively. Meanwhile, Jena et al. [17] present an AI framework focusing on similar challenges, and Sakly et al. [18] introduce a transfer learning-based approach to classify data, demonstrating the utility and adaptability of these advanced methods in addressing complex problems under resource constraints.

Addressing this problem using CNNs or RNNs individually presents certain limitations. CNNs struggle to efficiently capture temporal or sequential information, while RNNs may overlook important spatial discriminative features. Fusion techniques offer a potential solution but selecting the appropriate fusion method remains a challenge. Additionally, while transformers excel at capturing global information, they often struggle to focus on local features. Data scarcity further complicates fusion, as it may emphasize redundant information. In this work, we aim to tackle these challenges through a multi-fusion approach, combining transformers with CNN-RNN hybrids to balance both global and local feature extraction.

3. Methodology

The proposed methodology comprises three key phases. In the first phase, a cascade network is trained, utilizing both a convolutional neural network (CNN) and a transformer in parallel to leverage their complementary feature extraction capabilities. The second phase focuses on intermediate-level fusion, where intermediate feature maps are extracted using contrastive learning and subsequently fused to enhance the representation. Finally, in the decision-level fusion phase, the confidence scores of the logits are aligned to refine and finalize the model's predictions. Fig. 1 shows the overall flow of the proposed framework. Following is the description of all the phases:

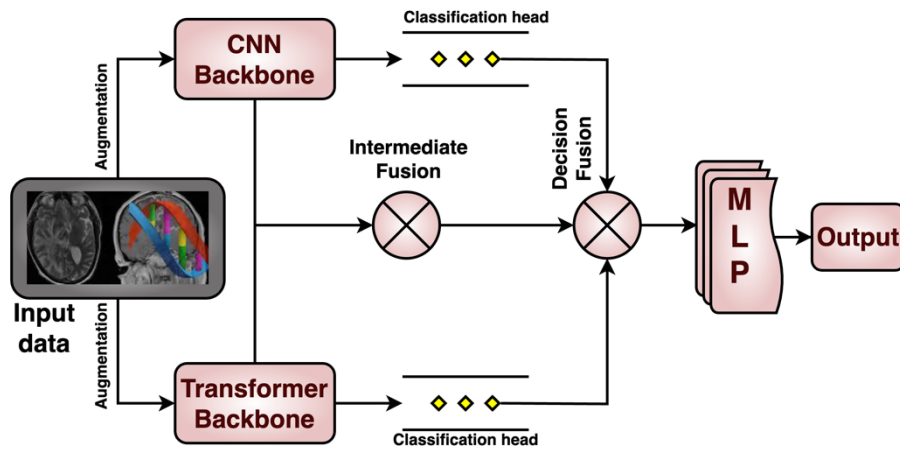


Fig. 1. The proposed framework that integrates CNN and transformer features at intermediate and decision levels.

3.1. CNN Transformer training:

In this phase, the dataset undergoes preprocessing and augmentation, as detailed in the implementation section. Let the preprocessed and augmented input data be denoted as x . The Convolutional Neural Network (CNN) function applied to the input is represented as $fc(\cdot)$, while the Transformer function applied is denoted by $ft(\cdot)$. Here,

$fc(x)=h_c$, where h_c denote the CNN feature vector.

$ft(x)=h_t$, where h_t denote the Transformer feature vector.

At this stage, the input undergoes transformation through the application of CNNs and Transformers. This dual transformation approach is designed to extract both local and global information. The CNN is employed to capture localized, spatially relevant features by focusing on nearby regions within the input, while the Transformer leverages self-attention mechanisms to model long-range dependencies and global relationships. As a result, we obtain both spatial features and the structural arrangement of these features through the Transformer. The transformed input is subsequently passed through the fusion mechanism, which we discuss in detail in the following section.

3.1.1. Intermediate Level Fusion

In this section, we describe the operations performed during intermediate-level fusion. At this stage, the feature vectors h_c and h_t are combined. The fusion can be carried out using various techniques such as concatenation, element-wise addition, or element-wise multiplication of normalized unit vectors. In our implementation, we opt for concatenation as the fusion strategy. The equation for the same is as follows:

$$h_F = \text{concat}(h_C, h_T). \quad (1)$$

Following this, the fused feature vector h_F , got from equation 1 is input into a classifier to generate the logits. We employ a two-layer Multi-Layer Perceptron (MLP) on top of the concatenated vector to compute the logits using the following equation:

$$z_F = f_{\text{cls}}(h_F). \quad (2)$$

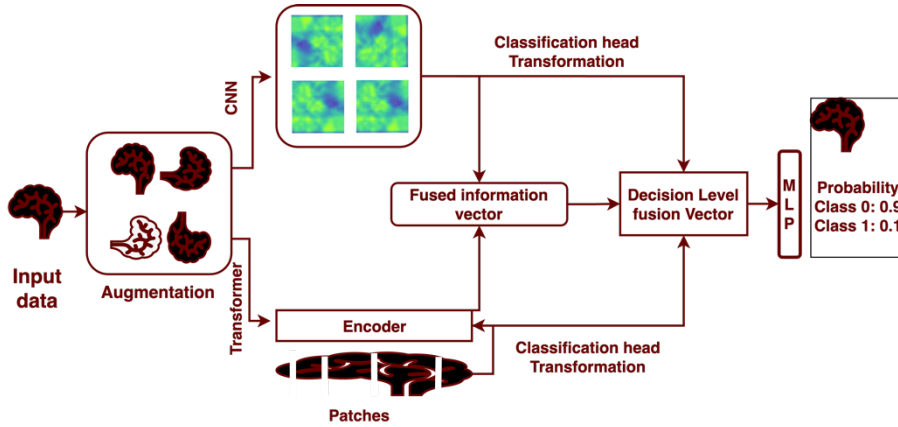


Fig. 2. Different phases of image transformation using the proposed framework

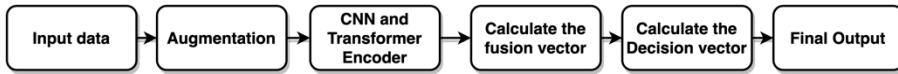


Fig. 3. Flowchart of the proposed framework

Next, in this study, we propose a decision-level fusion approach to regularize the stored data, thereby enhancing class discrimination in a structured manner. This methodology aims to improve the accuracy and robustness of classification outcomes by reinforcing the distinctions among different classes.

3.1.2. Decision Level Fusion

At the decision level, we compute the logits separately from the CNN and Transformer:

$$z_C = f_{\text{cls}}(h_C), \quad z_T = f_{\text{cls}}(h_T). \quad (3)$$

The logits are fused using a weighted sum based on the different kind of logits calculated from equations 2 and 3:

$$z_{\text{dec}} = \alpha z_C + \beta z_T + \gamma z_F. \quad (4)$$

Where, the sum of α , β and γ should be 1.

Finally, apply the MLP and softmax function to obtain the output probabilities p_F from the final logits got from equation 4:

$$p_F = \text{softmax}(M(z_{\text{dec}})), \quad (5)$$

Where M denotes transformation obtained after using the Multi-Layer Perceptron (MLP). Fig. 2 shows the overall transformation of the data from input to output, which is got from equation 1 to equation 5, whereas Fig. 3 shows the flowchart representing flow of the framework.

3.2. Contrastive Learning:

To achieve effective transformations through deep network learning, we employ contrastive self-supervised learning techniques. This approach leverages the similarity between anchor points and data samples to extract refined dark knowledge. By focusing on the relationships between positive and negative pairs, our method enhances the

model's ability to discern subtle features within the data, thereby facilitating improved representation learning and knowledge distillation. This process not only promotes a deeper understanding of the underlying data structures but also contributes to the overall robustness of the model.

3.3. Classification Task:

For the final classification task, we utilize a multilayer perceptron (MLP) to process the fused knowledge, generating the final probability outputs. The MLP is specifically designed to capture complex patterns and relationships within the integrated feature representations. It is trained using cross-entropy loss, which quantifies the difference between the predicted probabilities and the actual class labels. This loss function effectively guides the optimization process, ensuring that the MLP learns to produce accurate classifications by minimizing prediction errors across the training dataset. The combination of the MLP and the fused knowledge enhances the model's performance in distinguishing between classes.

4. Experimental Settings

In this section, we delve into the details of the experimental settings, including the dataset, implementation, hardware, and performance metrics.

4.1. The dataset

We consider RSNA-MICCAI Brain Tumor Radiogenomic Classification Challenge [2] dataset for the analysis. The dataset is structured into three distinct cohorts, comprising a total of 672 subjects, each carefully categorized based on their role in the model development process. Specifically, the dataset includes:

- 468 subjects for training
- 117 subjects for validation
- 87 subjects for testing

Each subject in the dataset is identified by a unique five-digit number. This numerical identifier is used to label and organize the subjects' data. Within each subject's folder, multiple types of multi-parametric MRI (mpMRI) scans are provided in DICOM format. These mpMRI scans offer a comprehensive view of the brain's structure and function, with each scan type capturing different anatomical or pathological features. The specific types of mpMRI scans included for each subject are:

- **T1-weighted pre-contrast (T1w):** This scan provides detailed anatomical information by highlighting the normal structures of the brain. T1w scans are typically used for assessing brain tissue integrity and identifying large tumors.
- **T1-weighted post-contrast (T1wCE):** After administering a contrast agent, this scan captures enhanced images that help visualize blood-brain barrier disruptions, making it useful for detecting tumors, particularly malignant ones, by highlighting regions of abnormal contrast uptake.
- **T2-weighted (T2w):** This scan captures high-resolution images that are sensitive to differences in water content, making it valuable for identifying lesions, swelling, or other abnormalities in the brain's soft tissues, including the presence of tumors.
- **Fluid Attenuated Inversion Recovery (FLAIR):** FLAIR scans suppress the signal from cerebrospinal fluid, making it easier to detect abnormalities like edema, demyelination, or small lesions near the ventricles. This sequence is particularly useful for identifying low-grade tumors and subtle brain abnormalities.

This comprehensive dataset, with its multi-modal imaging and structured cohorts, provides a rich foundation for developing deep learning models aimed at brain tumor detection and classification, while also ensuring rigorous evaluation through well-defined training, validation, and testing phases.

4.2. Implementation

The proposed framework is implemented using the PyTorch framework in Python. For training, we utilize energy-efficient Nvidia T4x2 GPUs with 16 GB of memory. Initially, we perform the required data preprocessing steps, followed by the application of our framework. The model is trained with a batch size of 32 and a learning rate of $1e-4$, incorporating a constant decay factor of 0.1 after 25 epochs. The training process spans 100 epochs, with optimization handled by the Adam optimizer. Additionally, we apply softmax non-linearity to the predictions to refine the output. For preprocessing, we follow the work in literature. For this, DICOM images were converted into 3D Numpy arrays representing height, width, and depth, and resized to 256x256 with 64 frames per image. A Value of Interest (VOI) Lookup Table was applied to enhance contrast for improved image quality. To increase dataset diversity, augmentation techniques such as 90-degree clockwise and counterclockwise rotations, as well as 180-degree rotations, were used. For training EfficientNet-B3, ResNet50, and Xception models, the RSNA MICCAI dataset from Kaggle was utilized. Images were resized to 300x300 for EfficientNet-B3, and to 512x512 for ResNet50 and Xception. Random augmentation, including horizontal flips, rotations ($\pm 36^\circ$), and random horizontal/vertical translations ($\pm 10\%$), was applied to prevent overfitting during training.

4.3. Network architectures

We consider several models based on literature recommendations: ResNet, known for its residual learning and deep architecture; Xception, which leverages depthwise separable convolutions for efficiency; EfficientNet B3 offering a scalable architecture balancing accuracy and efficiency; and MobileNet, optimized for lightweight, high-performance models suitable for mobile and embedded applications.

4.4. Performance metrics

The performance metrics used in this study include accuracy, F1-score, precision, and recall. Accuracy is selected for its importance in classification tasks, providing an overall measure of correctness. However, given the dataset's inherent data scarcity, it's crucial to evaluate other aspects as well. Precision and recall are employed to assess the model's efficiency in terms of true positive and true negative rates, ensuring a more comprehensive understanding of the model's performance. Additionally, the F1-score is used to provide a balanced measure that considers both precision and recall, offering a more robust evaluation of the model's effectiveness.

5. Results and discussions

To analyze the efficiency of the proposed framework, baseline methods reported in the literature [1] are compared against its application. Performance metrics, including accuracy, F1-score, and AUC, are calculated and compared to assess the effectiveness of the framework. Table 1 and 2 provide such comparison.

Table 1. Performance metrics for classification.

Model	Accuracy	F1-Score	Sensitivity	Specificity
Resnet50	0.47	0.37	0.30	0.67
EfficientnetB3	0.40	0.32	0.27	0.56
Xception	0.60	0.50	0.42	0.80
Proposed Resnet50-ViT	0.49	0.38	0.31	0.64
Proposed EfficientnetB3-ViT	0.41	0.30	0.29	0.55
Proposed Xception-ViT	0.61	0.53	0.40	0.80

Table 2. AUC results for comparison.

Model	Validation AUC	Test AUC
Resnet50	0.41	0.57
EfficientnetB3	0.37	0.55
Xception	0.63	0.61
Proposed Resnet50-ViT	0.41	0.55
Proposed EfficientnetB3-ViT	0.37	0.54
Proposed Xception-ViT	0.64	0.63

The comparison reveals that the baseline methods, particularly XceptionNet, demonstrate good performance. However, after applying our proposed framework, the networks achieve results that are either comparable to or exceed those of existing approaches. For example, XceptionNet shows an approximate 3% improvement in F1-score compared to standard training settings. This highlights the effectiveness of multilevel fusion combined with contrastive learning for this type of classification task. The improved accuracy demonstrates the method's enhanced performance on the data. Table 1 also indicates high sensitivity and specificity, contributing to greater robustness in handling data scarcity and class imbalance. Table 2 indicates that the AUROC is often high which reflects a higher ability to correctly classify positive and negative instances, even under varying decision thresholds, demonstrating robustness and reliability in predictions. Next, we delve into the ablation study for the same.

5.1. Ablation study

In this section, we analyze the performance of individual components of the proposed framework to determine which part contributes the most. For this, we utilize ViT-32 as the vision transformer based on the literature [1] and evaluate the performance of its individual components sequentially. The study demonstrates that ViTs and Xception, when used independently, are unable to effectively extract dark knowledge, resulting in suboptimal performance as presented in Table 3. However, with the implementation of our proposed framework, these models exhibit improved performance, which shows that the amalgamation of both backbones (Xception and ViT) positively affects the ability of the proposed method. We will now delve into the key applications and challenges of the study.

Table 3. Ablation study for comparison.

Model	Validation AUC	Test AUC
Xception	0.63	0.61
VIT	0.58	0.60
Proposed Xception-ViT	0.64	0.63

5.2. Key Applications and Challenges

The proposed method is particularly suited for addressing challenges like data scarcity and class imbalance, with applications in medical imaging and other data-driven fields. A significant challenge encountered was ensuring enhanced integration of local and global features while preserving the utility of dark knowledge. Moreover, achieving effective fusion at both intermediate and decision levels required precise calibration of CNN and transformer contributions, combined with the use of contrastive learning to strengthen feature representation. The key conclusions of this study are summarized next.

6. Conclusion

In this article, we propose a novel approach for dark knowledge extraction and efficient fusion using a multi-step fusion framework. Our method extracts dark knowledge from both CNN and transformer architectures in parallel, fusing it at the intermediate level for enhanced knowledge integration and at the logit level for precise outcome alignment. The inclusion of contrastive learning further boosts performance. This approach yields a 3-4% performance

improvement across various studies and proves effective in different backbone networks. As a future scope, we can explore advanced transfer learning mechanisms and adaptive weighting to refine fusion and enhance scalability to other domains like multi-modal and 3D medical imaging. Incorporating self-supervised learning and generative models can further improve performance in data-scarce scenarios.

Acknowledgements

I express my gratitude to GLA University for providing the research infrastructure. Special thanks to Dr. Rakesh Galav (GLA University) and Mr. Saurabh Sharma (IIT Patna) for their invaluable advice, encouragement, and support throughout this work.

References

- [1] Mohamed A, Rabea M, Sameh A, Kamal E. Brain Tumor Radiogenomic Classification. arXiv preprint arXiv:2401.09471. 2024 Jan 11.
- [2] Baid U, Ghodasara S, Mohan S, Bilello M, Calabrese E, Colak E, Farahani K, Kalpathy-Cramer J, Kitamura FC, Pati S, Prevedello LM. The rsna-asnr-miccai brats 2021 benchmark on brain tumor segmentation and radiogenomic classification. arXiv preprint arXiv:2107.02314. 2021 Jul 5.
- [3] Kollias D, Vandal K, Gadhavi P, Russom S. Btdnet: A multi-modal approach for brain tumor radiogenomic classification. Applied Sciences. 2023 Nov 2;13(21):11984.
- [4] Sharma S, Kumar A, Chandra J. Confidence Matters: Enhancing Medical Image Classification Through Uncertainty-Driven Contrastive Self-distillation. In International Conference on Medical Image Computing and Computer-Assisted Intervention 2024 Oct 3 (pp. 133-142). Cham: Springer Nature Switzerland.
- [5] Sharma S, Kumar A, Monpara J, Chandra J. StAIK: Structural Alignment based Self Knowledge distillation for Medical Image Classification. Knowledge-Based Systems. 2024 Sep 11:112503.
- [6] Sharma S, Lodhi SS, Chandra J. SCL-IKD: intermediate knowledge distillation via supervised contrastive representation learning. Applied Intelligence. 2023 Dec;53(23):28520-41.
- [7] Tiwari AK, Sharma S, Kumar D. Hybrid Approach For Brain Tumor Segmentation, Dogo Rangsang Research Journal, 2020 Jul.
- [8] Sharma S, Tiwari AK, Kumar D. Identification of Emotions from Sentences Using Natural Language Processing for Small Dataset. Advances in Applications of Data-Driven Computing. 2021:35-47.
- [9] Abdusalomov AB, Mukhiddinov M, Whangbo TK. Brain tumor detection based on deep learning approaches and magnetic resonance imaging. Cancers. 2023 Aug 18;15(16):4172.
- [10] Saeedi S, Rezayi S, Keshavarz H, R. Niakan Kalhori S. MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques. BMC Medical Informatics and Decision Making. 2023 Jan 23;23(1):16.
- [11] Yogananda CG, Shah BR, Nalawade SS, Murugesan GK, Yu FF, Pinho MC, Wagner BC, Mickey B, Patel TR, Fei B, Madhuranthakam AJ. MRI-based deep-learning method for determining glioma MGMT promoter methylation status. American Journal of Neuroradiology. 2021 May 1;42(5):845-52.
- [12] Qureshi SA, Hussain L, Ibrar U, Alabdulkreem E, Nour MK, Alqahtani MS, Nafie FM, Mohamed A, Mohammed GP, Duong TQ. Radiogenomic classification for MGMT promoter methylation status using multi-omics fused feature space for least invasive diagnosis through mpMRI scans. Scientific reports. 2023 Feb 25;13(1):3291.
- [13] Patel SK, Singh A. Task Scheduling in Cloud Computing Using Hybrid Meta-Heuristic: A Review. In Proceedings of the International Conference on Paradigms of Computing, Communication and Data Sciences: PCCDS 2020 2021 (pp. 453-472). Springer Singapore.
- [14] Liu W, Nie S, Yin J, Wang R, Gao D, Jin L. SSKD: Self-supervised knowledge distillation for cross domain adaptive person re-identification. In 2021 7th IEEE International Conference on Network Intelligence and Digital Content (IC-NIDC) 2021 Nov 17 (pp. 81-85). IEEE.
- [15] Zhang P, Li Y, Wang D, Wang J. RS-SSKD: Self-supervision equipped with knowledge distillation for few-shot remote sensing scene classification. Sensors. 2021 Feb 24;21(5):1566.
- [16] Qu R, Xiao Z. An attentive multi-modal CNN for brain tumor radiogenomic classification. Information. 2022 Mar 2;13(3):124.
- [17] Jena B, Saxena S, Nayak GK, Balestrieri A, Gupta N, Khanna NN, Laird JR, Kalra MK, Fouda MM, Saba L, Suri JS. Brain tumor characterization using radiogenomics in artificial intelligence framework. Cancers. 2022 Aug 22;14(16):4052.
- [18] Sakly H, Said M, Seekins J, Guetari R, Kraiem N, Marzougui M. Brain Tumor Radiogenomic Classification of O6-Methylguanine-DNA Methyltransferase Promoter Methylation in Malignant Gliomas-Based Transfer Learning. Cancer Control. 2023 Apr 3;30:10732748231169149.
- [19] Raina S, Khandelwal A, Gupta S, Leekha A. Brain tumor segmentation using Unet. In Futuristic Trends in Network and Communication Technologies: Third International Conference, FTNCT 2020, Taganrog, Russia, October 14–16, 2020, Revised Selected Papers, Part I 3 2021 (pp. 432-443). Springer Singapore.
- [20] Magomedov SG. Software for analyzing security for healthcare organizations. In Futuristic Trends in Network and Communication Technologies: Third International Conference, FTNCT 2020, Taganrog, Russia, October 14–16, 2020, Revised Selected Papers, Part I 3 2021 (pp. 181-189). Springer Singapore.